

3.3 Matrix Multiplication: The Third Perspective



Core skills you'll master in this section:

- ▶ The column-vector-linear-combination perspective: for the matrix multiplication AB , each column vector of the result is a weighted combination of A 's column vectors.
- ▶ Ab is exactly A 's column vectors combined, weighted by the coefficients in b , into a new column vector.
- ▶ In the matrix multiplication AB , B 's every column serves as a weight applied to A 's every column; the weighted sum generates C 's every column.
- ▶ In the matrix multiplication AB , B has however many columns, AB has that many columns.

The third perspective on matrix multiplication is a way of viewing matrix multiplication as a linear combination of column vectors. This perspective, starting from the column-vector point of view, reveals another essential nature of matrix multiplication — for the matrix multiplication $C = A @ B$, C 's every column is the result of a linear combination of A 's column vectors.

Let's first look at $A @ b = c$

Matrix A has shape $m \times p$, and column vector b has shape $p \times 1$; their product $Ab = c$ has shape $m \times 1$.

Figure 1 shows the first perspective on the matrix multiplication $Ab = c$.

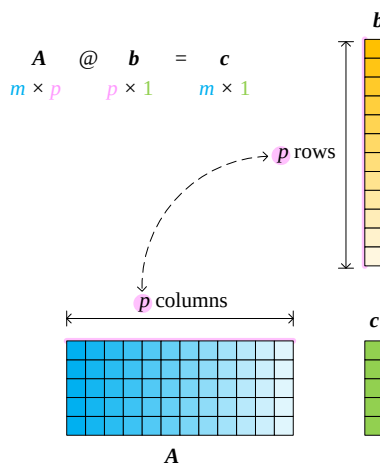


Figure 1. Matrix Multiplication $Ab = c$

Linear Combination of Column Vectors

Write $\mathbf{A}$ as a set of column vectors

$$\mathbf{A}_{m \times p} = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_p]$$

Among these, column vector $\mathbf{a}_k$ ($k = 1, 2, \dots, p$) has shape $m \times 1$.

The matrix multiplication $\mathbf{A} @ \mathbf{b}$ can be written as

$$\mathbf{A}_{m \times p} \mathbf{b}_{p \times 1} = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_p] \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_p \end{bmatrix} = b_1 \mathbf{a}_1 + b_2 \mathbf{a}_2 + \cdots + b_p \mathbf{a}_p = \mathbf{c}$$

As shown in Figure 2, $\mathbf{c}$ is a linear combination (weighted sum) of the $\mathbf{A}$'s p column vectors. We call this the third perspective on matrix multiplication — the column-vector-linear-combination perspective.

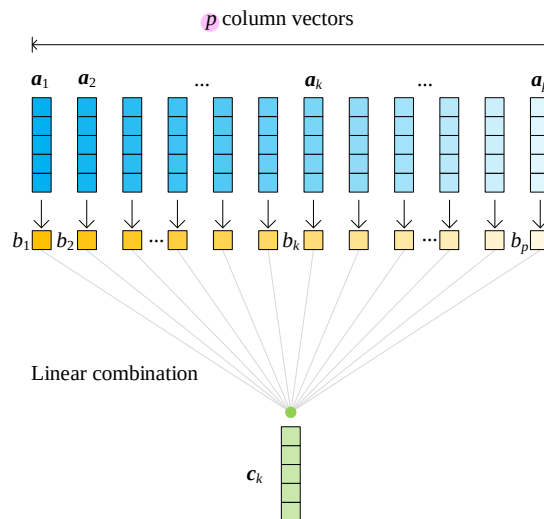


Figure 2. $\mathbf{A}$'s Linear Combination of Column Vectors

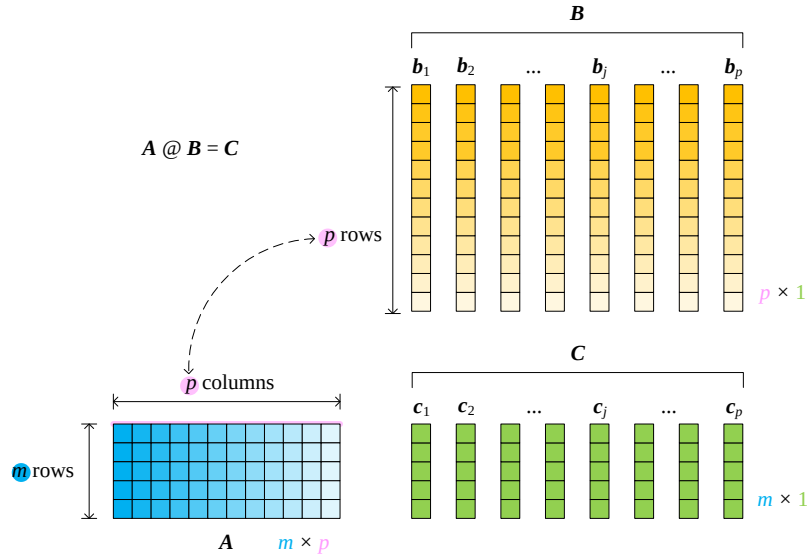
Looking Again at $\mathbf{A} @ \mathbf{B} = \mathbf{C}$

Similarly, we can also use the column-vector-linear-combination perspective to look at the matrix multiplication $\mathbf{C} = \mathbf{A} @ \mathbf{B}$.

As shown in Figure 3, write matrices $\mathbf{C}$ and $\mathbf{B}$ respectively as sets of column vectors; then the matrix multiplication $\mathbf{C} = \mathbf{A} @ \mathbf{B}$ can be written as

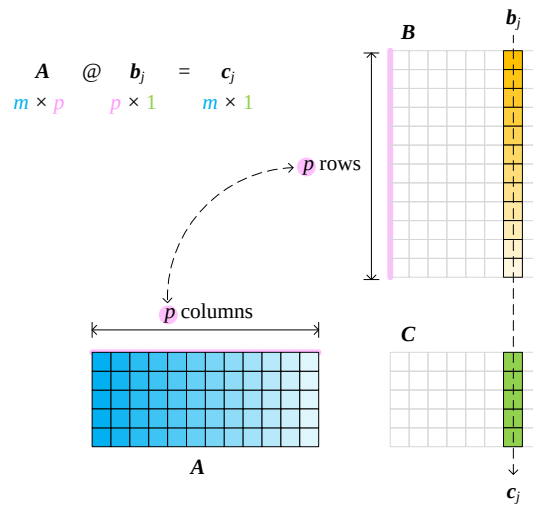
$$\mathbf{C} = [\mathbf{c}_1 \quad \mathbf{c}_2 \quad \cdots \quad \mathbf{c}_n] = \mathbf{A} @ \mathbf{B} = \mathbf{A} @ [\mathbf{b}_1 \quad \mathbf{b}_2 \quad \cdots \quad \mathbf{b}_n] = [\mathbf{A} @ \mathbf{b}_1 \quad \mathbf{A} @ \mathbf{b}_2 \quad \cdots \quad \mathbf{A} @ \mathbf{b}_n]$$

Figure 3 also tells us why matrix $\mathbf{B}$ has p columns, and why the matrix multiplication $\mathbf{A} @ \mathbf{B}$ also has p columns.

Figure 3. Write Matrices C and B Respectively as Sets of Column Vectors

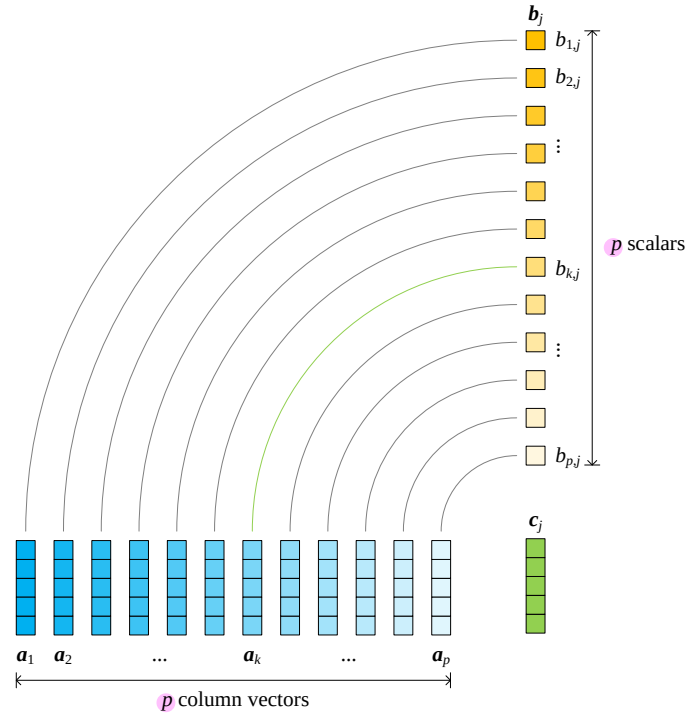
As shown in Figure 4, any column of matrix C can be computed via the formula below

$$c_j = A @ b_j$$

Figure 4. $c_j = A b_j$ – the Corresponding Matrix Multiplication

Therefore, as shown in Figure 5, matrix C 's j -th column vector c_j can be further written as a linear combination of the column vectors of matrix A , with the combination coefficients provided by matrix B 's j -th column vector b_j 's components, that is

$$c_j = A @ b_j = \begin{bmatrix} a_1 & a_2 & \cdots & a_p \end{bmatrix} \begin{bmatrix} b_{1,j} \\ b_{2,j} \\ \vdots \\ b_{p,j} \end{bmatrix} = b_{1,j} a_1 + b_{2,j} a_2 + \cdots + b_{p,j} a_p$$

Figure 5. Matrix $\mathbf{A}$'s Column-Vector-Linear-Combination Perspective on $\mathbf{c}_j = \mathbf{A}\mathbf{b}_j$

Conversely, after separately computing $\mathbf{A}\mathbf{b}_1, \mathbf{A}\mathbf{b}_2, \mathbf{A}\mathbf{b}_3$, and so on, arranging them in order into a row gives matrix $\mathbf{C}$ that is

$$[\mathbf{A}@\mathbf{b}_1 \quad \mathbf{A}@\mathbf{b}_2 \quad \dots \quad \mathbf{A}@\mathbf{b}_n] = [\mathbf{c}_1 \quad \mathbf{c}_2 \quad \dots \quad \mathbf{c}_n]$$

The First Example

Let's look at the matrix-multiplication example covered earlier in this book; first, symbolically expand $\mathbf{A}@\mathbf{b}_1$

$$\mathbf{c}_1 = \mathbf{A}@\mathbf{b}_1 = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3] @ \begin{bmatrix} b_{1,1} \\ b_{2,1} \\ b_{3,1} \end{bmatrix} = b_{1,1} \cdot \mathbf{a}_1 + b_{2,1} \cdot \mathbf{a}_2 + b_{3,1} \cdot \mathbf{a}_3$$

Substituting specific values, compute $\mathbf{c}_1$

$$\mathbf{c}_1 = \mathbf{A}@\mathbf{b}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} @ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 1 \times \begin{bmatrix} 1 \\ 4 \\ 6 \end{bmatrix} + 2 \times \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} + 3 \times \begin{bmatrix} 3 \\ 6 \\ 18 \end{bmatrix} = \begin{bmatrix} 1+4+9 \\ 4+10+18 \\ 32 \end{bmatrix} = \begin{bmatrix} 14 \\ 32 \end{bmatrix}$$

Symbolically expand $\mathbf{A}@\mathbf{b}_2$

$$\mathbf{c}_2 = \mathbf{A}@\mathbf{b}_2 = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3] @ \begin{bmatrix} b_{1,2} \\ b_{2,2} \\ b_{3,2} \end{bmatrix} = b_{1,2} \cdot \mathbf{a}_1 + b_{2,2} \cdot \mathbf{a}_2 + b_{3,2} \cdot \mathbf{a}_3$$

Substituting specific values, compute $\mathbf{c}_2$

$$\mathbf{c}_2 = \mathbf{A} @ \mathbf{b}_2 = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} @ \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = 4 \times \begin{bmatrix} 1 \\ 4 \end{bmatrix} + 5 \times \begin{bmatrix} 2 \\ 5 \end{bmatrix} + 6 \times \begin{bmatrix} 3 \\ 6 \end{bmatrix} = \begin{bmatrix} 4+10+18 \\ 16+25+36 \end{bmatrix} = \begin{bmatrix} 32 \\ 77 \end{bmatrix}$$

$$\begin{array}{l} b_{1,1} \times \frac{a_1}{a_1} + b_{2,1} \times \frac{a_2}{a_2} + b_{3,1} \times \frac{a_3}{a_3} = c_1 \\ 1 \times \frac{1}{4} + 2 \times \frac{2}{5} + 3 \times \frac{3}{6} = \frac{14}{32} \\ \\ b_{1,2} \times \frac{a_1}{a_1} + b_{2,2} \times \frac{a_2}{a_2} + b_{3,2} \times \frac{a_3}{a_3} = c_2 \\ 4 \times \frac{1}{4} + 5 \times \frac{2}{5} + 6 \times \frac{3}{6} = \frac{32}{77} \end{array}$$

Figure 6. Computing $\mathbf{c}_1$ and $\mathbf{c}_2$

As shown in Figure 7, then arrange the column vectors $\mathbf{c}_1$ and $\mathbf{c}_2$ in order to get matrix $\mathbf{C}$

$$\mathbf{C} = \mathbf{A} @ \mathbf{B} = [\mathbf{c}_1 \quad \mathbf{c}_2] = \begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix}$$

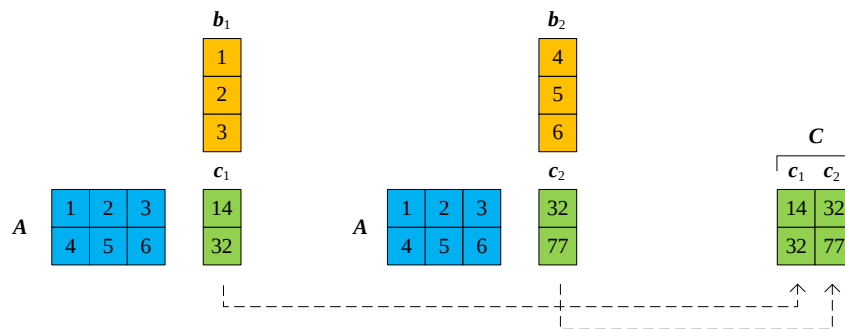


Figure 7. Arranging Column Vectors $\mathbf{c}_1$ and $\mathbf{c}_2$ in Order to Get Matrix $\mathbf{C}$

The Second Example

Let's use the linear-combination perspective to look at the second matrix-multiplication example. Substituting specific values, compute $\mathbf{d}_1$

$$\mathbf{d}_1 = \mathbf{B} @ \mathbf{a}_1 = 1 \times \mathbf{b}_1 + 4 \times \mathbf{b}_2 = 1 \times \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 4 \times \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1+16 \\ 2+20 \\ 3+24 \end{bmatrix} = \begin{bmatrix} 17 \\ 22 \\ 27 \end{bmatrix}$$

Compute $\mathbf{d}_2$

$$\mathbf{d}_2 = \mathbf{B} @ \mathbf{a}_2 = 2 \times \mathbf{b}_1 + 5 \times \mathbf{b}_2 = 2 \times \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 5 \times \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 2+20 \\ 4+25 \\ 6+30 \end{bmatrix} = \begin{bmatrix} 22 \\ 29 \\ 36 \end{bmatrix}$$

Compute d_3

$$d_3 = B @ a_3 = 3 \times b_1 + 6 \times b_2 = 3 \times \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 6 \times \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 3+24 \\ 6+30 \\ 9+36 \end{bmatrix} = \begin{bmatrix} 27 \\ 36 \\ 45 \end{bmatrix}$$

$$\begin{array}{c} a_{1,1} \\ 1 \end{array} \times \begin{array}{c} b_1 \\ 1 \\ 2 \\ 3 \end{array} + \begin{array}{c} a_{2,1} \\ 4 \end{array} \times \begin{array}{c} b_2 \\ 4 \\ 5 \\ 6 \end{array} = \begin{array}{c} d_1 \\ 17 \\ 22 \\ 27 \end{array}$$

$$\begin{array}{c} a_{1,2} \\ 2 \end{array} \times \begin{array}{c} b_1 \\ 1 \\ 2 \\ 3 \end{array} + \begin{array}{c} a_{2,2} \\ 5 \end{array} \times \begin{array}{c} b_2 \\ 4 \\ 5 \\ 6 \end{array} = \begin{array}{c} d_2 \\ 22 \\ 29 \\ 36 \end{array}$$

$$\begin{array}{c} a_{1,3} \\ 3 \end{array} \times \begin{array}{c} b_1 \\ 1 \\ 2 \\ 3 \end{array} + \begin{array}{c} a_{2,3} \\ 6 \end{array} \times \begin{array}{c} b_2 \\ 4 \\ 5 \\ 6 \end{array} = \begin{array}{c} d_3 \\ 27 \\ 36 \\ 45 \end{array}$$

Figure 8. Computing d_1 , d_2 , and d_3

As shown in Figure 9, arrange d_1 , d_2 , and d_3 side by side to get matrix D

$$D = B @ A = [d_1 \ d_2 \ d_3] = \begin{bmatrix} 17 & 22 & 27 \\ 22 & 29 & 36 \\ 27 & 36 & 45 \end{bmatrix} = \begin{bmatrix} 17 & 22 & 27 \\ 22 & 29 & 36 \\ 27 & 36 & 45 \end{bmatrix}$$

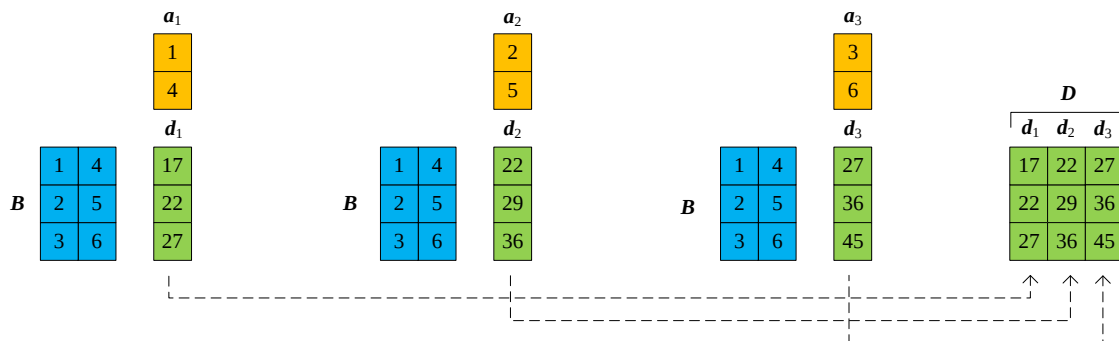


Figure 9. Arranging Column Vectors d_1 , d_2 , and d_3 in Order to Get Matrix D



LA_03_03_01.ipynb completes the third-perspective matrix-multiplication computations above – please study it on your own.

Visualizing the Third Perspective on Matrix Multiplication with a Heatmap

Next, let's use a heatmap to visualize the third perspective on matrix multiplication.

Figure 10 uses a heatmap to show how to compute $A @ b_1$.

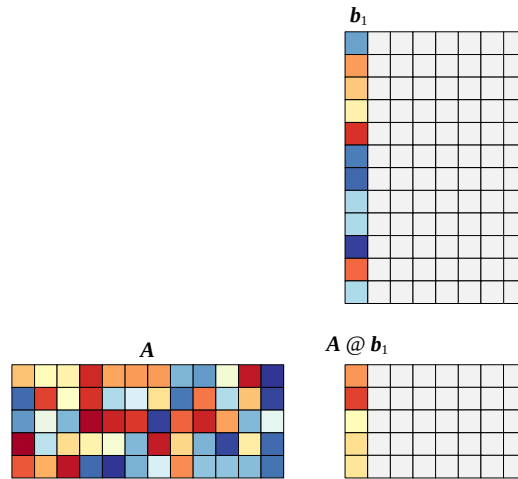


Figure 10. Heatmap Showing $A @ b_1$

Since B has 8 columns, there are matrix multiplications like $A @ b_1$, 8 in total; the results are shown in Figure 11. Arranging their results in order gives matrix $C = A @ B$.

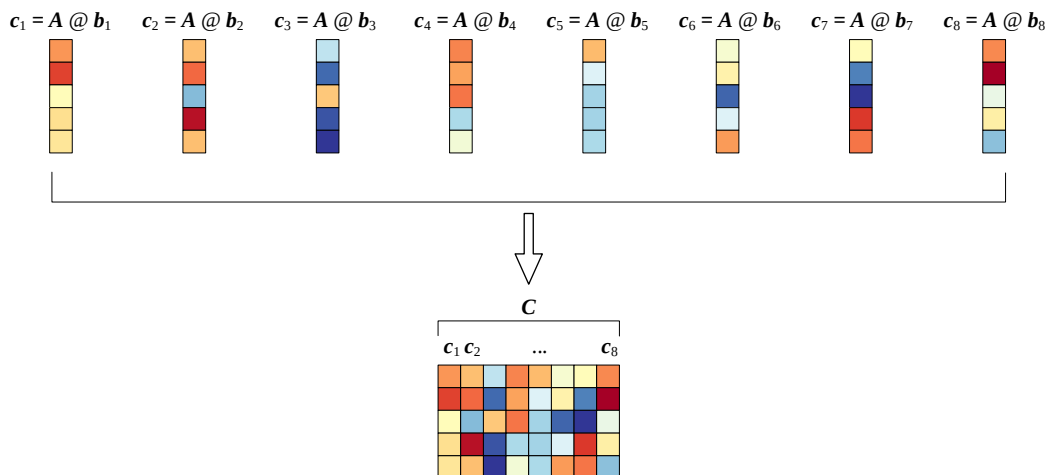


Figure 11. 8 $A @ b_j$ Side by Side Give Matrix $C = A @ B$



LA_03_03_02.ipynb uses `seaborn.heatmap()` to draw the heatmap above, visualizing the third perspective on matrix multiplication — please study it on your own.

A Custom Python Function to Compute Matrix Multiplication: The Column-Vector-Linear-Combination Perspective

Code 1 uses a custom function to complete the matrix multiplication using the column-vector-linear-combination perspective. Let's go over the key statements below.

a Defines a function called `col_LC`, meaning “column-vector linear combination.” A function is a piece of code we can reuse. This function takes two inputs: a matrix A and a coefficient list `coefficients`; it ultimately returns a vector representing each column of the matrix combined using these coefficients.

b Starts a loop, from column 0 through column $p-1$. `A[:, k]` extracts the k -th column of A ; then we multiply this column vector by the corresponding coefficient `coefficients[k]`, and add it to the running result. Repeating this several times, we eventually get the result of the linear combination of column vectors.

c We then define a new function, `matrix_multiplication_col_LC`, whose job is to implement “matrix multiplication” — but in a special way: through “column-vector combination.” The inputs are two matrices A and B , and it ultimately returns a new matrix C , representing A times B 's result.

d Starts a loop, from column 0 through column $n-1$, meaning we compute the result matrix C 's columns, one at a time. `C[:, j] = col_LC(A, coeffs)` calls the `col_LC()` function defined earlier, passing in A and the coefficients for the current column. The function multiplies every column of A by its corresponding coefficient and sums them; the result is the new matrix C 's j -th column.

Code 1. A Custom Python Function to Compute Matrix Multiplication, the Column-Vector-Linear-Combination Perspective | 
LA_03_03_03.ipynb


```

## 初始化
import numpy as np

## 定义列向量线性组合函数
def col_LC(A, coefficients):

    m, p = A.shape
    col_LC_result = np.zeros(m) # 初始化组合结果

    for k in range(p):
        col_LC_result += coefficients[k] * A[:, k]
        # A (左侧) 每列乘系数并相加
    return col_LC_result

## 定义矩阵乘法函数, 列向量组合
def matrix_multiplication_col_LC(A, B):
    # 获取矩阵 A 和 B 的形状
    m, p_A = A.shape
    p_B, n = B.shape

    # 检测矩阵形状是否符合矩阵乘法规则
    if p_A != p_B:
        raise ValueError('Dimensions do not match')

    # 初始化结果矩阵 C, 形状 (m, n), 初始值设为 0
    C = np.zeros((m, n))

    for j in range(n):
        coeffs = B[:, j]
        # 取出 B 的第 j 列, 作为线性组合的系数
        C[:, j] = col_LC(A, coeffs)
        # 计算 A 的列的线性组合

    return C

## 矩阵乘法

A = np.array([[1, 2, 3],
               [4, 5, 6]])
B = A.T

## 矩阵乘法
matrix_multiplication_col_LC(A, B)
matrix_multiplication_col_LC(B, A)

```



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Use the column-vector-linear-combination method to compute the following pairs of matrix multiplications

This PDF file is the author's draft, published to make it convenient for readers to study on mobile devices; the final content is subject to the printed edition published by Tsinghua University Press.

Copyright belongs to Tsinghua University Press. Please do not use it commercially, and please credit the source when quoting.
Code and PDF file downloads: <https://github.com/Visualize-ML>

The companion video lectures for this book are all published on Bilibili — 生姜 DrGinger: <https://space.bilibili.com/513194466>
Comments and corrections are welcome. The book's dedicated email address: jiang.visualize.ml@gmail.com

► $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

► $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

► $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

► $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

Q2. Use the third perspective on matrix multiplication to expand the following pairs of matrix multiplications

► $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

► $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

► $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = A^T$

► $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = A^T$

Q3. Combine the following two matrix multiplications into one.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} @ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} @ \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Q4. Combine the following three matrix multiplications into one, and give the result.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} @ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} @ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Q5. Combine the following three matrix multiplications into one, and give the result.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} @ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} @ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Q6. Compare the results of Q4 and Q5, and explain the similarities and differences between them.

Q7. Please study permutation matrices on your own.

Q8. A chicken has one head and two feet; a rabbit has one head and four feet. A cage currently holds 5 chickens and 8 rabbits — use matrix multiplication to compute the total number of heads and feet in the cage.